ch4 Gas Law and Kinetic Theory

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2022 Q3

*3. A weather balloon filled with 21 kg of helium gas has a volume of 120 m^3 at 27 °C. Find the gas pressure in the balloon. Given: mass of one mole of helium gas = 4×10^{-3} kg

A. 9.93×10^4 Pa
B. 1.00×10^5 Pa
C. 1.05×10^5 Pa
D. 1.09×10^5 Pa

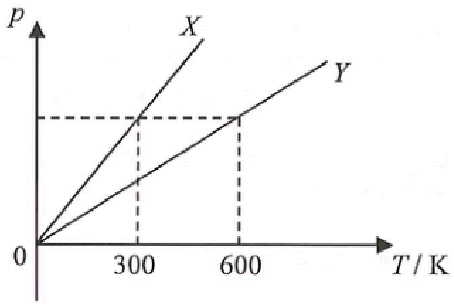
2021 Q2

2. Metal blocks X and Y are identical in size and shape. X is of a higher temperature than Y. Which of the following statements must be correct ?

(1) Energy will flow from X to Y if they are in thermal contact.
(2) X is a better conductor of heat compared to Y.
(3) The total internal energy of X is higher than that of Y.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

*3. A fixed mass of an ideal gas contained in a closed vessel X of volume V is heated. Line X in the figure shows the variation of pressure p with temperature T of the gas. The experiment is repeated with another closed vessel Y containing the same amount of this ideal gas and line Y represents the result.



Find the volume of vessel Y.

A. 0.25 V
B. 0.5 V
C. 2V
D. 4V

2017 Q4

*4. The pressure of a fixed mass of an ideal gas at 10°C is 2×10^5 N m^{-2}. If the volume of the gas is reduced to half of its original volume and its temperature is increased to 100°C, what would the pressure be ?

A. 1.00×10^5 N m^{-2}
B. 1.32×10^5 N m^{-2}
C. 4.00×10^5 N m^{-2}
D. 5.27×10^5 N m^{-2}

2015 Q3

3. When an object P is in contact with another object Q, heat flows from P to Q. P must have a higher

(1) temperature.
(2) internal energy.
(3) specific heat capacity.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

2017 Q3

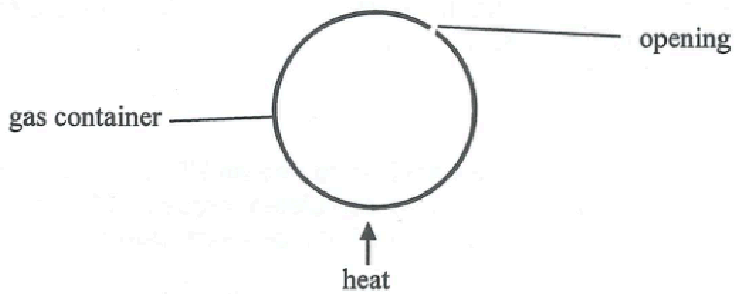
3. Which of the following statements about the internal energy of a substance are correct ?

(1) When a solid melts, the latent heat of fusion absorbed becomes potential energy of the molecules in the substance.
(2) When a vapour condenses, its internal energy decreases.
(3) When a liquid evaporates, the internal energy of the remaining liquid increases.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2019 Q3

*3. The figure shows an inexpansible container with an opening.



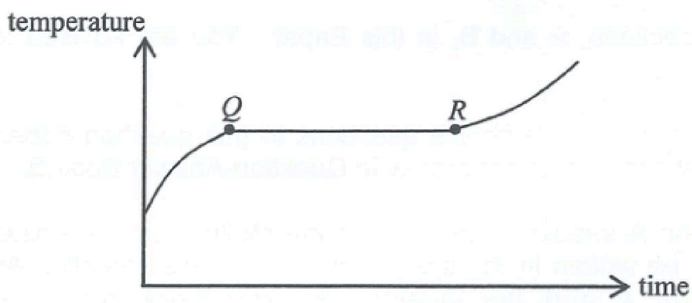
When the gas inside the container is being heated slowly by a heater, which statements about the gas molecules in the container are correct ?

(1) The number of molecules decreases.
(2) The average kinetic energy of molecules increases.
(3) The average separation between molecules remains unchanged.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2019 Q2

2.



A substance undergoes the fusion process. The figure shows how the substance's temperature varies with time. Its temperature remains constant during the period from Q to R. Which of the following deductions **within this period** is/are correct ?

(1) The substance does not absorb heat.
(2) The mass ratio of the substance in solid and liquid states remains constant throughout.
(3) The average potential energy of the molecules of the substance increases with time.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2022 Q2

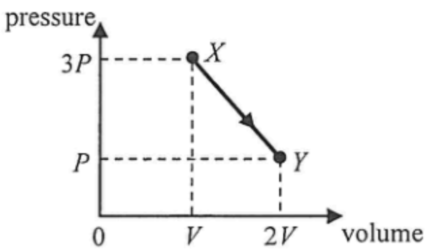
2. When ice melts into water at 0 °C, what happens to the molecules during the melting process ?

(1) their average separation increases
(2) their average kinetic energy increases
(3) their average potential energy increases

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2020 Q3

*3. A fixed mass of an ideal gas expands from state X to state Y through a process as represented in the pressure-volume graph below.



If the temperature of the gas at state Y is 25 °C, what is its temperature at state X ?

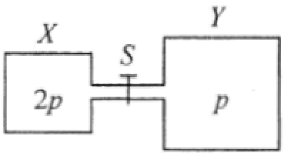
A. −74.3 °C
B. 16.7 °C
C. 37.5 °C
D. 174 °C

2024 Q3

*3. The best vacuum attained in the laboratory has a pressure of about 10^{-8} Pa. Estimate the order of magnitude of the number of air molecules in 1 cm^3 of such 'vacuum' at room temperature.

A. 10^4
B. 10^6
C. 10^8
D. 10^{12}

*4.



Vessel X of volume V and vessel Y of volume $2V$ are connected by a short narrow tube as shown. Initially, tap S is closed and the same kind of ideal gas at the same temperature is contained in X and Y at pressure $2p$ and p respectively. The tap S is then opened and equilibrium state is finally reached with the temperature unchanged. Which statement is **INCORRECT** ?

A. Before S is opened, both vessels contain the same number of gas molecules.

B. Before S is opened, the average kinetic energy of the gas molecules in both vessels is the same.

C. When S is opened, a net flow of gas from X to Y occurs.

D. When equilibrium is reached, the gas pressure becomes $\frac{3}{2}p$.

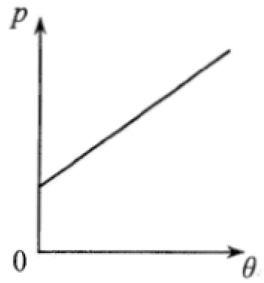
2016 Q3

*3. When an ideal gas is heated from 25 °C to 50 °C, the average kinetic energy of the gas molecules will

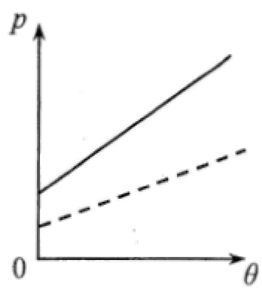
A. double.
B. increase by 41 %.
C. increase by 8.4%.
D. increase by 4.1%.

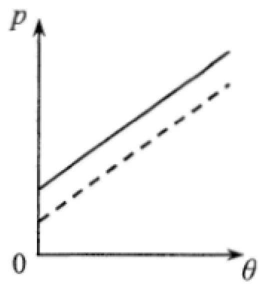
2012 Q3

*3. An ideal gas is contained in a closed vessel of fixed volume. The graph below shows the variation of pressure p of the gas against its Celsius temperature θ.



If the number of gas molecules in the vessel is halved, which graph of the dotted line best shows the relationship between p and θ ?

A.


B.


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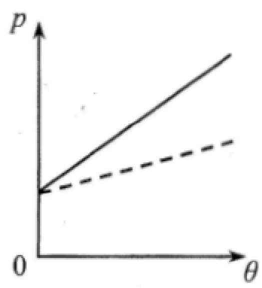

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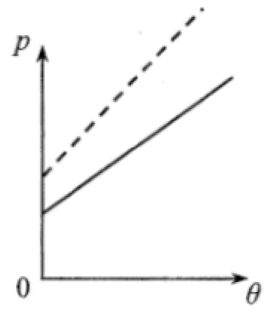


*3. Identical closed vessels X and Y contain carbon dioxide (CO_2) and oxygen (O_2) respectively.

Both gases are at room temperature and atmospheric pressure. Which of the following physical quantities of the gas molecules in the two vessels is/are the same ?

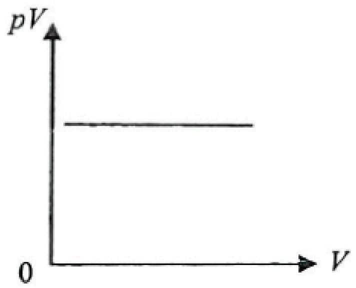
(1) root-mean-square speed
(2) average kinetic energy
(3) frequency of collision with the walls of the vessel

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

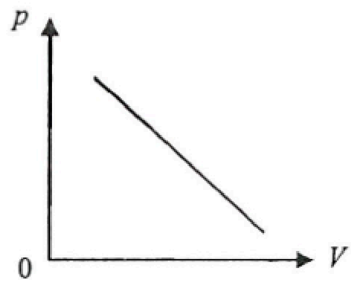
2018 Q4

*4. From which graph below can one deduce that the pressure p of a fixed mass of an ideal gas is inverse proportional to its volume V when the temperature of the gas is kept constant ?

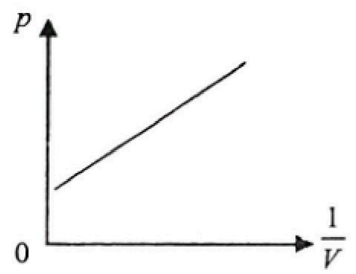
A.



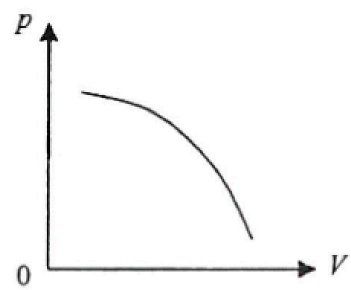
B.



C.



D.



2013 Q3

*3. In which of the following situations would the r.m.s. speed of the molecules of a fixed mass of an ideal gas increase ?

(1) The gas is heated under constant volume.
(2) The gas expands under constant pressure.
(3) The gas is compressed under constant temperature.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

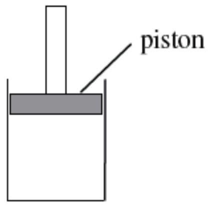
2021 Q3

*3. For an ideal gas, kinetic theory deduces that $pV = \frac{1}{3}Nm\overline{c^2}$. Which physical quantity below can be represented by $\frac{3p}{\overline{c^2}}$?

A. the total mass of the gas
B. the volume of one mole of the gas
C. the density of the gas
D. the number of gas molecules per unit volume

pp Q5

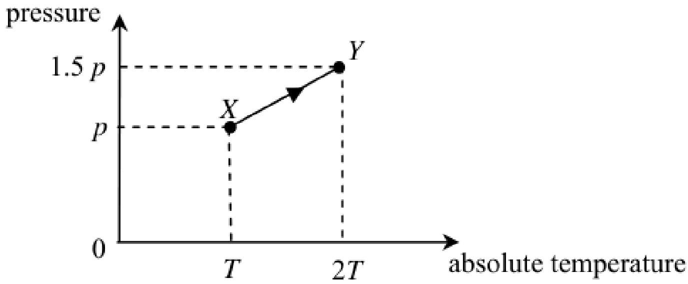
*5. A fixed mass of an ideal gas is contained in a cylinder fitted with a frictionless piston as shown in the figure below. If the gas is cooled under constant pressure,



(1) the average separation of the gas molecules will decrease.
(2) the r.m.s. speed of the gas molecules will decrease.
(3) the number of collisions per second of the gas molecules on the piston will decrease.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

*3.



As the gas in a vessel of fixed volume is heated, it gradually leaks out. The gas in the vessel changes from state X to state Y along the path XY shown in the plot of pressure against absolute temperature. What percentage of the original mass of the gas leaks out from the vessel in this process ?

A. 10%
B. 20%
C. 25%
D. 50%

sap Q4

*4. Two vessels contain hydrogen gas and oxygen gas respectively. Both gases have the same pressure and temperature and are assumed to be ideal. Which of the following physical quantities must be the same for the two gases ?

A. The volume of the gas
B. The mass per unit volume of the gas
C. The r.m.s. speed of the gas molecules
D. The number of gas molecules per unit volume